04 - Preprocessing and Feature Engineering

Customer Churn Prediction | Repository-based interview study guide

Source: https://github.com/YuvarajGarla/Customer-churn-prediction

1. Convert TotalCharges to numeric

```
telco_data.TotalCharges = pd.to_numeric(
    telco_data.TotalCharges, errors='coerce'
)
```

The errors=coerce option converts values that cannot be parsed into numbers into NaN. The notebook then finds 11 missing TotalCharges entries, which are removed using dropna. This is about 0.16% of the original 7,043 rows.

Important: A careful analysis should inspect those 11 rows. In this dataset, blank TotalCharges entries are associated with zero tenure. Dropping them is a small row loss, but the decision and its business/data implications should be documented.

2. Create tenure groups

```
labels = [f"{i} - {i+11}" for i in range(1, 72, 12)]
telco_data["tenure_group"] = pd.cut(
    telco_data.tenure, range(1, 80, 12),
    right=False, labels=labels
)
```

Group	Months
1 - 12	1 to 12
13 - 24	13 to 24
25 - 36	25 to 36
37 - 48	37 to 48
49 - 60	49 to 60
61 - 72	61 to 72

Binning can capture lifecycle stages and simplify interpretation. The trade-off is information loss: two customers with different tenure values in the same bin become indistinguishable with respect to this feature.

3. Remove identifier and original tenure

```
telco_data.drop(columns=['customerID', 'tenure'], inplace=True)
```

customerID is an identifier, not a generalizable behavioural feature. tenure is removed because tenure_group was created from it. This is a modelling choice, not a universal rule: keeping both could be tested if the model and validation design justify it.

4. Encode target

```
telco_data['Churn'] = np.where(
    telco_data.Churn == 'Yes', 1, 0
)
```

5. One-hot encode categorical features

```
telco_data_dummies = pd.get_dummies(telco_data)
```

One-hot encoding creates a separate binary indicator for each category. This avoids imposing an artificial numerical ordering on nominal categories. The repository applies pandas get_dummies to the prepared dataframe.

Important: For a robust production/reusable training workflow, use a fitted preprocessing transformer such as OneHotEncoder(handle_unknown="ignore") inside a Pipeline. Fit encoders on training data only and apply the same learned mapping to validation/test/new data. The Flask app's separate get_dummies logic may be sensitive to missing or unexpected categories and column order.

6. Feature scaling

The repository does not scale features before Decision Tree or Random Forest training. This is generally acceptable for tree-based methods because monotonic scaling of a feature preserves its ordering and possible threshold partitions. Scaling is important for many distance- or gradient-based algorithms and becomes relevant for PCA.

7. Interview-ready explanation

During preprocessing, I converted TotalCharges from object to numeric. This exposed 11 missing observations, which the notebook removed. I created 12-month tenure groups to represent customer lifecycle stages, removed customerID and the original tenure column, encoded Churn as 0/1, and applied one-hot encoding to categorical variables. The tree-based models used in the project do not require feature scaling.

8. Questions to prepare

Q. What does errors="coerce" do?

It converts unparseable values to NaN so they can be identified and handled explicitly.

Q. What is one-hot encoding?

It represents each category with a binary indicator column, avoiding an artificial ordinal relationship between nominal categories.

Q. Why might we not scale for Random Forest?

Tree splits depend on feature ordering and thresholds. A monotonic rescaling generally preserves the ordering and candidate partitions.

Q. What is the drawback of binning tenure?

It discards within-bin variation and can create arbitrary boundaries. It should be justified and compared against retaining continuous tenure.